

Task 2.1

This task contains dendrograms showing the results of different hierarchical clustering methods (single, complete, average, ward) on unscaled and scaled weather data from all stations between 2010 and 2020, all stations from 2015, Belgrade station between 2010 and 2020 and Valentia station between 2010 and 2020. The dendrograms are recreated each time after reducing the number of components in the datasets from all locations to 15, 5 and 2 components using PCA (principal component analysis), and reducing the components in the datasets from individual locations to 3, 2 and 1 component.

Each scaled and unscaled dataset contains a pair-plot comparing the weather data to Pleasant Weather answers from the corresponding time period. For the datasets taken from all weather stations, pair-plots were created only for the PCA reduced data, as the full data would have been too large to show in this kind of plot.

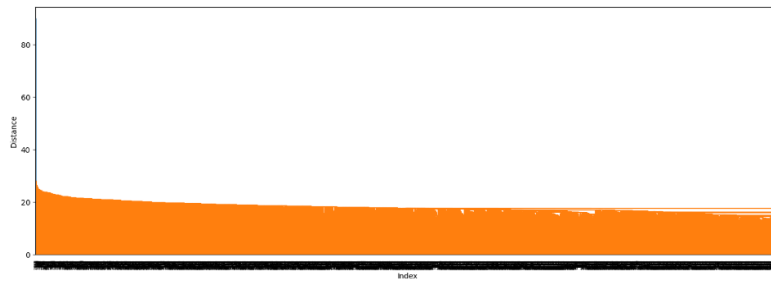
The complete set of dendrograms and pair-plots for the unscaled and scaled 2010-2020 data from all stations is shown in this document. For the rest of the dendrograms and pair-plots, please see the attached Jupyter notebooks.

Weather Data from All Stations, 2010-2020 (Unscaled)

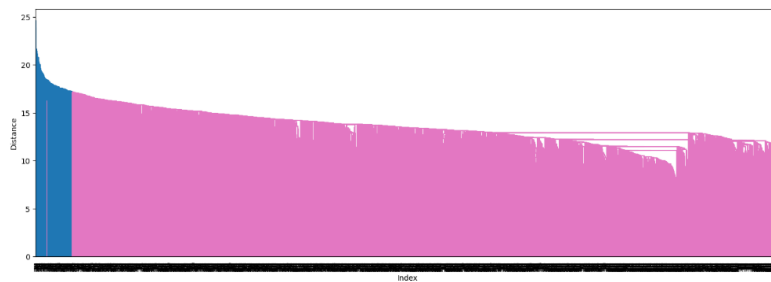
Single Method

- Determines the number of clusters based on the distance between the two closest points.
- Formed a long, single cluster of data points every time, even after applying PCA

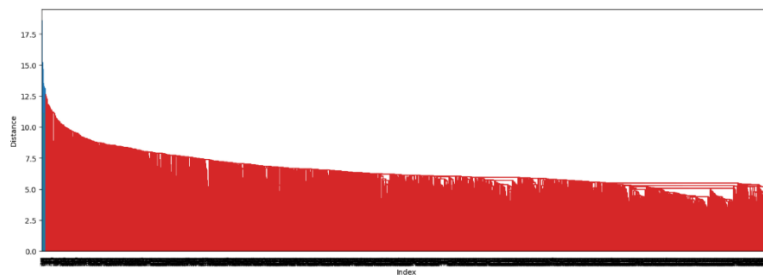
Dendrogram Single Method - All Weather Stations



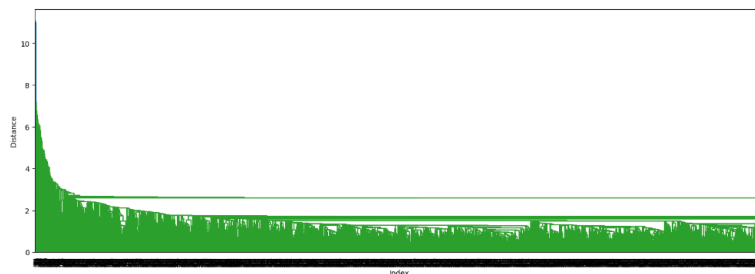
Dendrogram Single Method - All Weather Stations (Reduced Dimensions)



Dendrogram Single Method - All Weather Stations (Reduced to 5 Components)

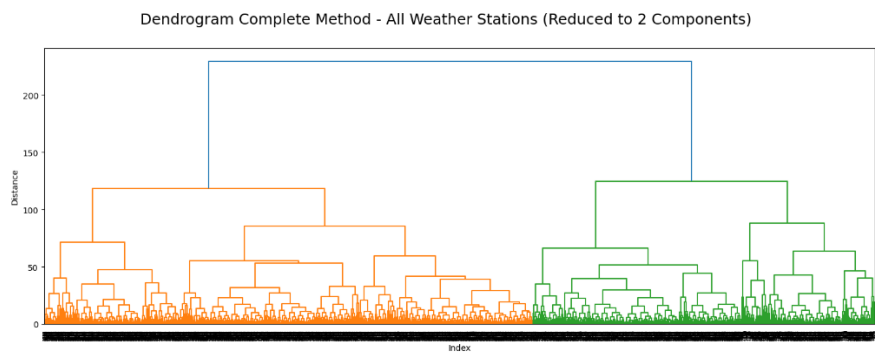
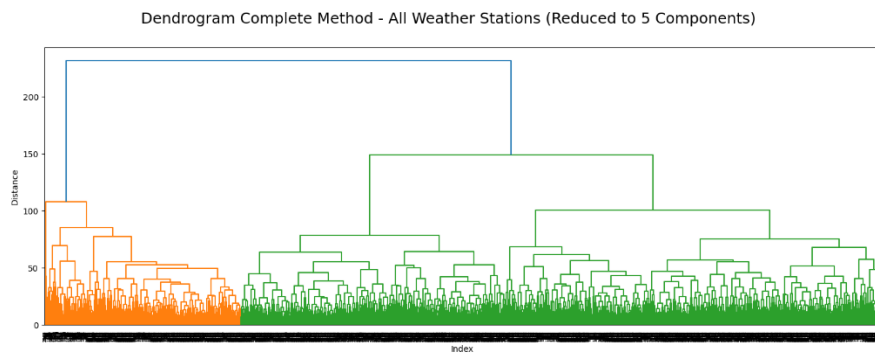
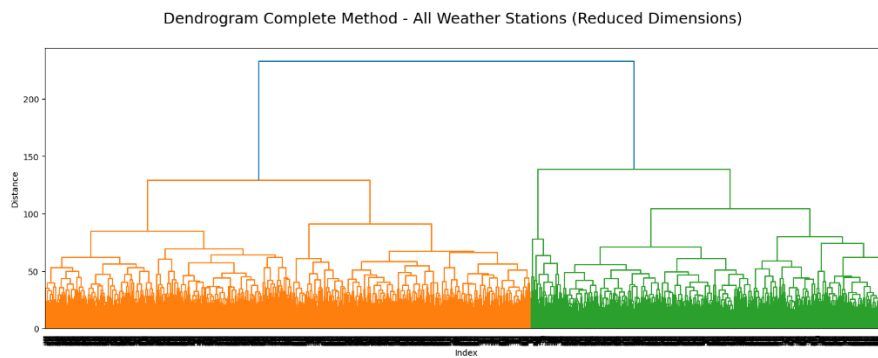
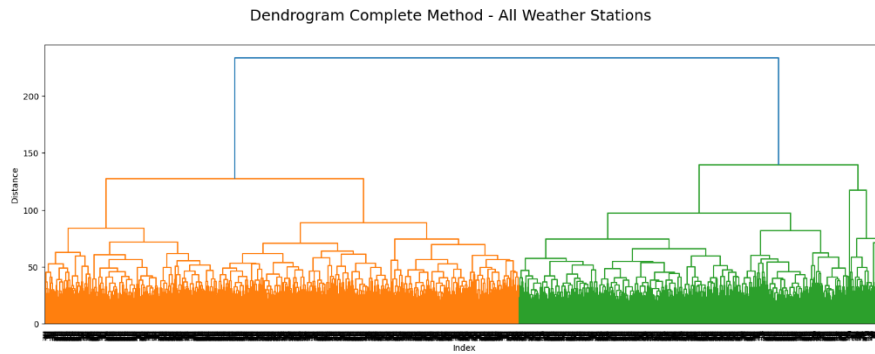


Dendrogram Single Method - All Weather Stations (Reduced to 2 Components)



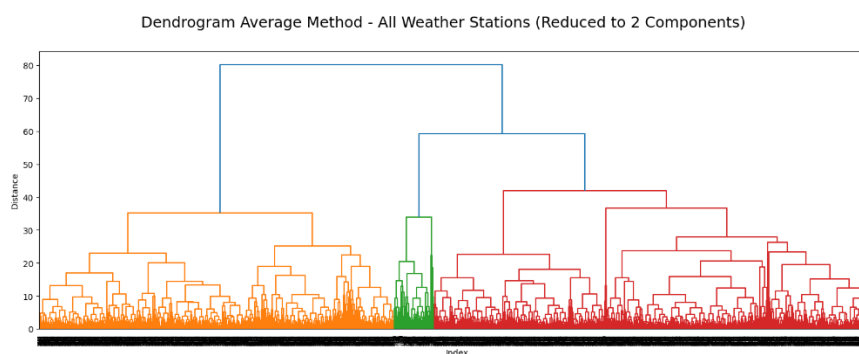
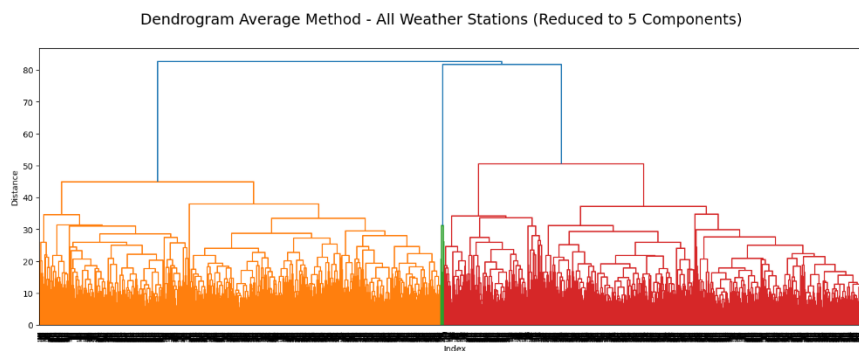
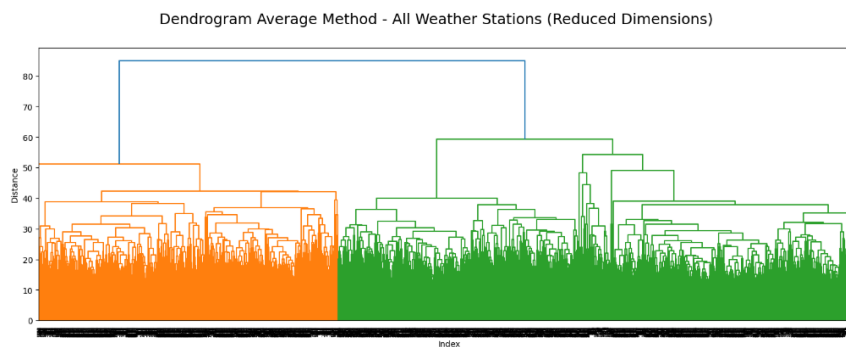
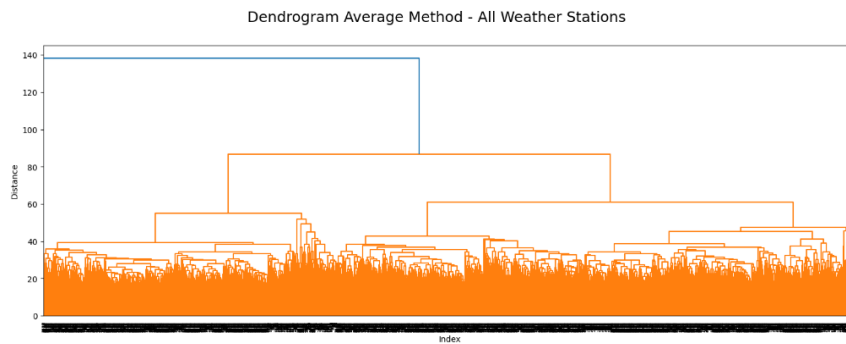
Complete Method

- Determines clusters based on the longest distance between two points.
- Formed two large, distinct clusters with a large number of nested clusters each.
- These clusters became clearer as the number of components decreased (15, 5 and 2).



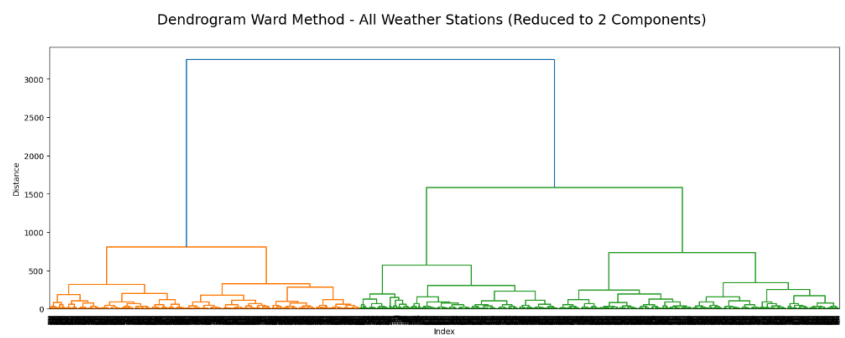
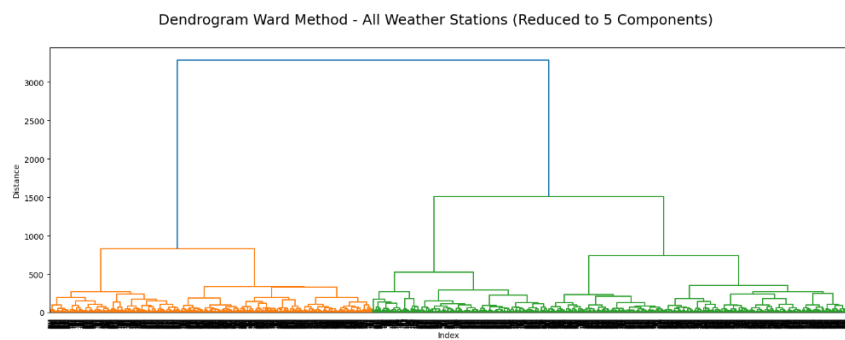
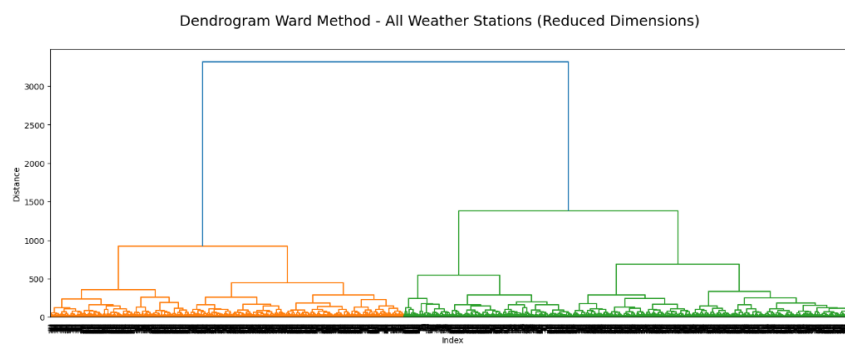
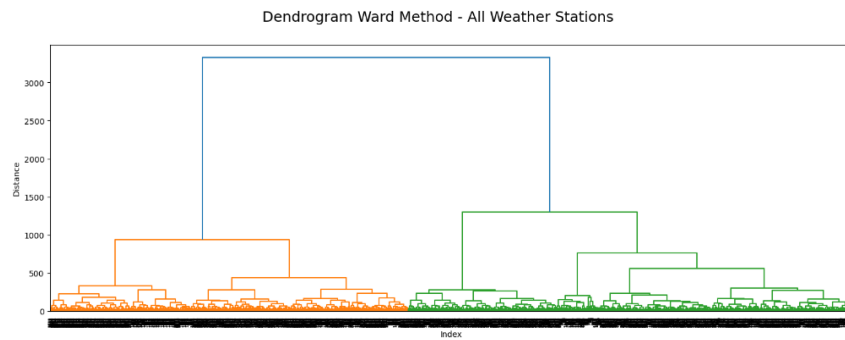
Average Method

- Determines clusters based on the distance between averages of each cluster.
- The number of clusters formed with this method increased as the number of components decreased (one large cluster with all of the data, 2 clusters with 15 components, 2 clusters plus one singleton with 5 components, 3 distinct clusters with 2 components).



Ward Method

- Uses the MISSQ (minimum increase of sum of squares) to find distance between clusters while minimizing variance.
- PCA component reduction did not appear to affect the number or size of the clusters formed using this method.



Pair-Plots with Pleasant Weather Data

Pair-plots showing the relationship between the 15 components of the weather data after the first round of PCA and the total number of Pleasant Weather answers per day.

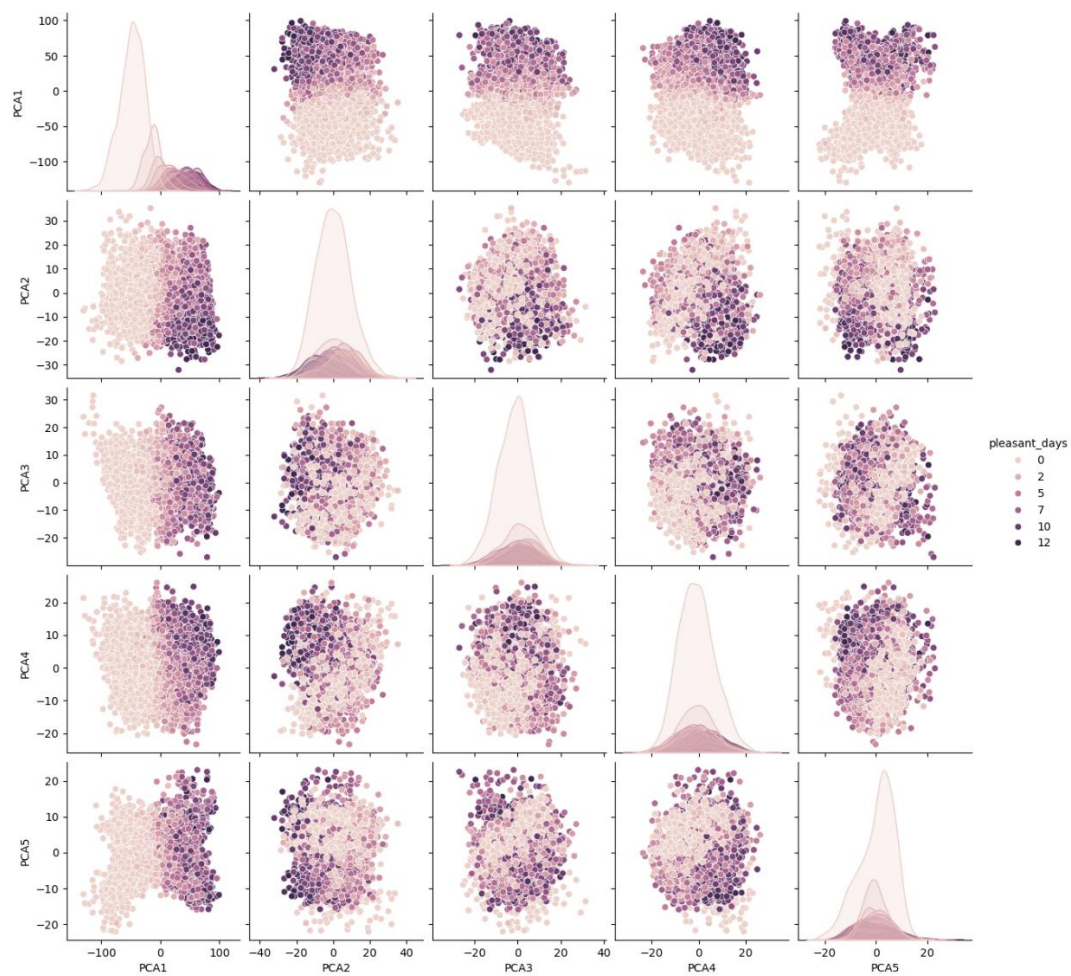
Observations

- All pair-plots showed the effect of certain components influencing the number of pleasant weather answers recorded each day.
- Even when the number of components was reduced to 5 and 2, there was a clear distinction between clusters with higher and lower counts of pleasant weather days.

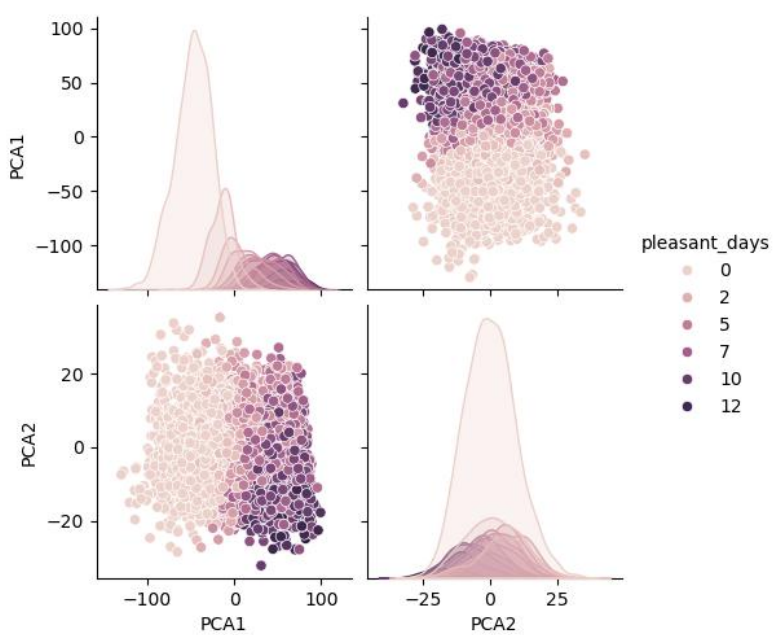
PCA Reduced (15 Components)



PCA Reduced (5 Components)



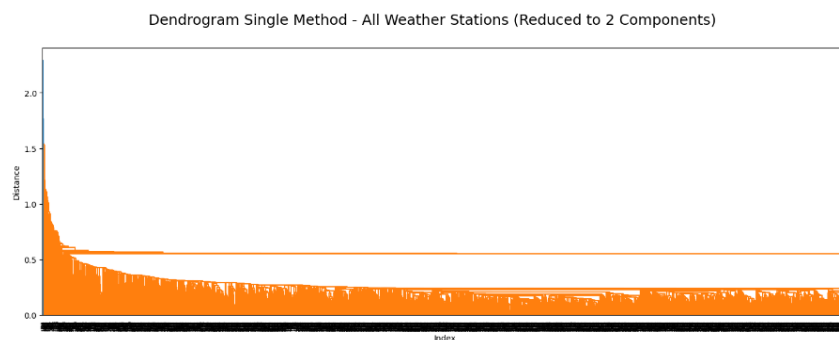
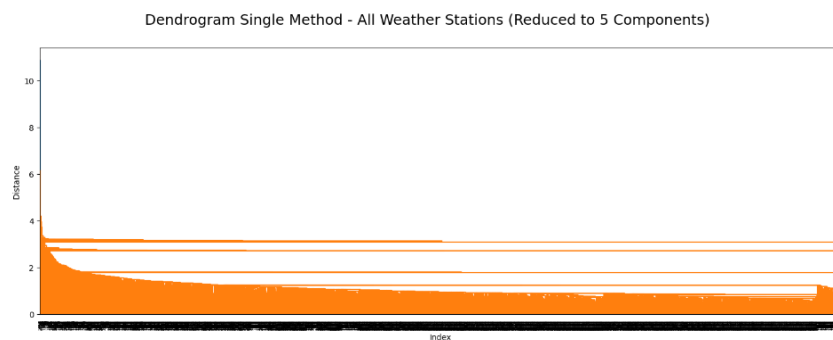
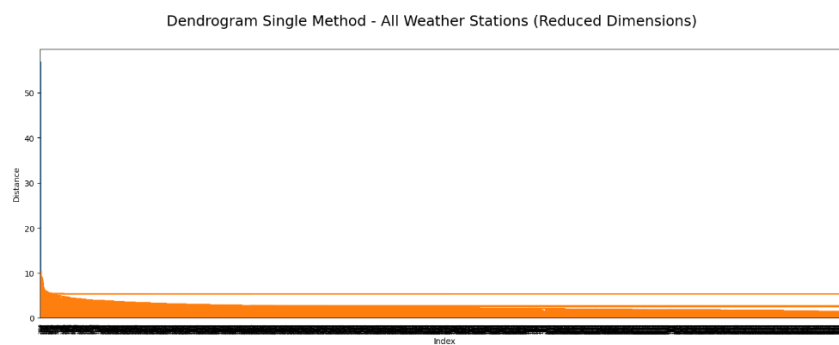
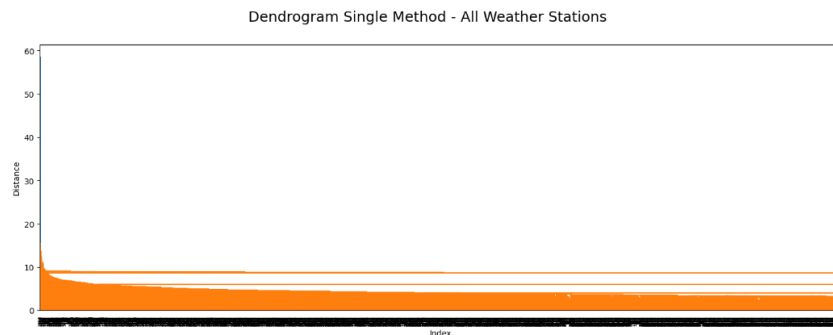
PCA Reduced (2 Components)



Weather Data from All Stations, 2010-2020 (Scaled)

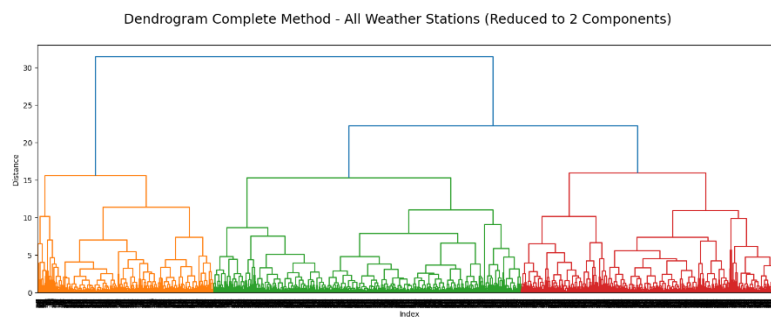
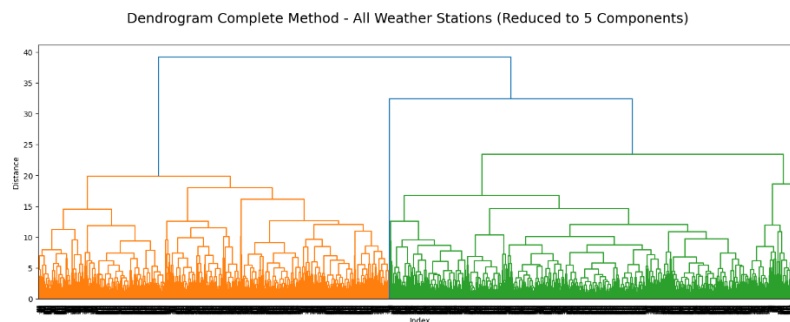
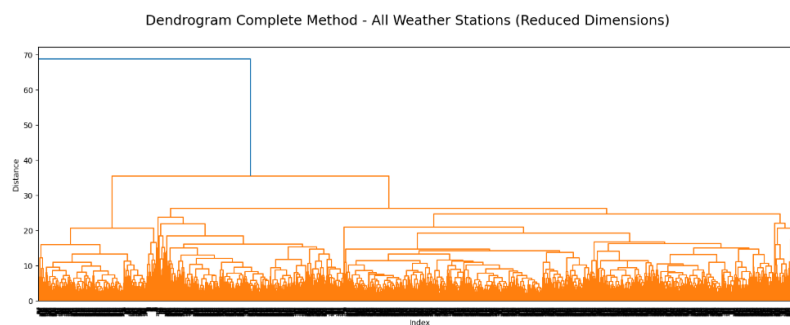
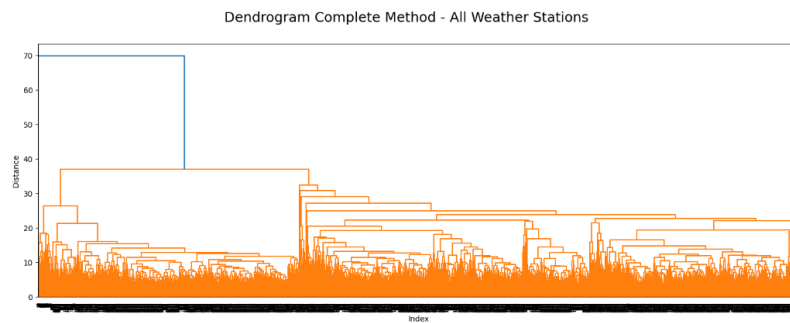
Single Method

- The Single method produces only one cluster with scaled data and is likely not very useful in classifying this data set. The shape of this single cluster remains similar throughout the PCA dimensionality reductions, resembling a long chain.
- Unlike with the unscaled data, the single-method dendrograms produced with the scaled data are all one colour.



Complete Method

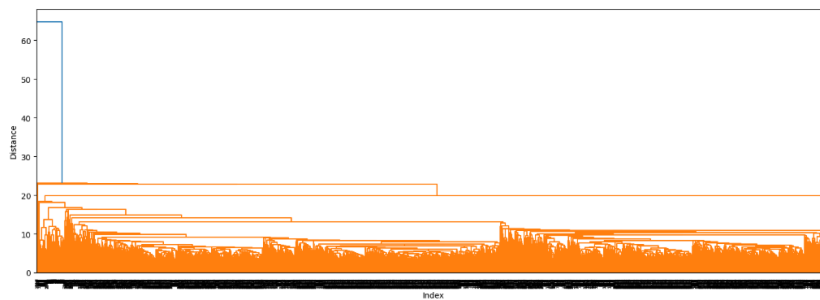
- The Complete method produces a higher number of clusters when the data is reduced to fewer components.
- Using all the data, as well as 15 components, this method produces one large cluster with several layers. With 5 components, we can see 2 distinct clusters and what appears to be third, singleton cluster in the middle. This third cluster increases in size with 2 components, resulting in 3 distinct clusters.



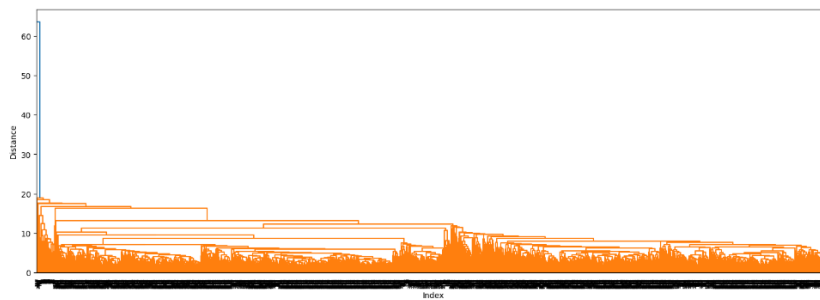
Average Method

- This method produces a single large cluster that appears to gradually split as the number of components is reduced. With the data reduced to 2 components, a very small, second cluster appears next to the larger one.
- This second cluster could represent extreme weather or some other relatively-uncommon weather occurrence.

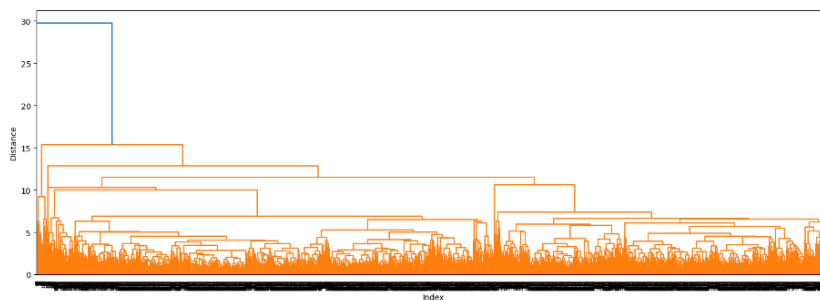
Dendrogram Average Method - All Weather Stations



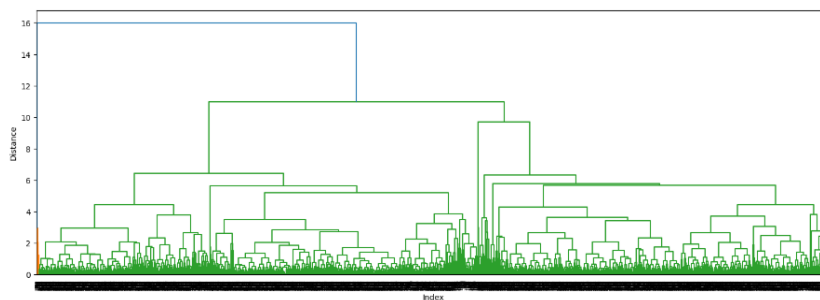
Dendrogram Average Method - All Weather Stations (Reduced Dimensions)



Dendrogram Average Method - All Weather Stations (Reduced to 5 Components)

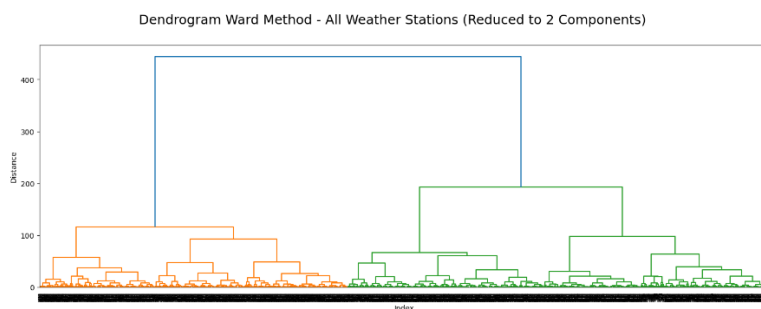
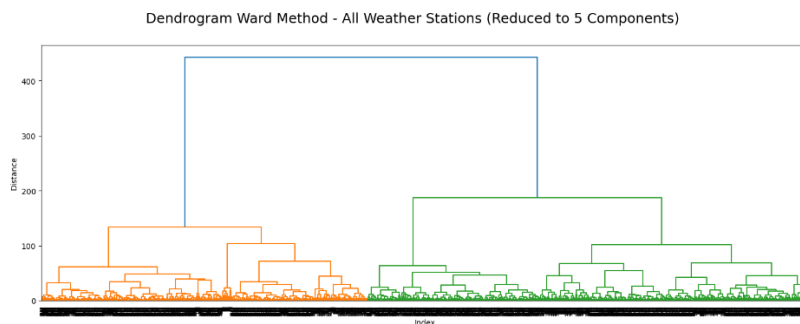
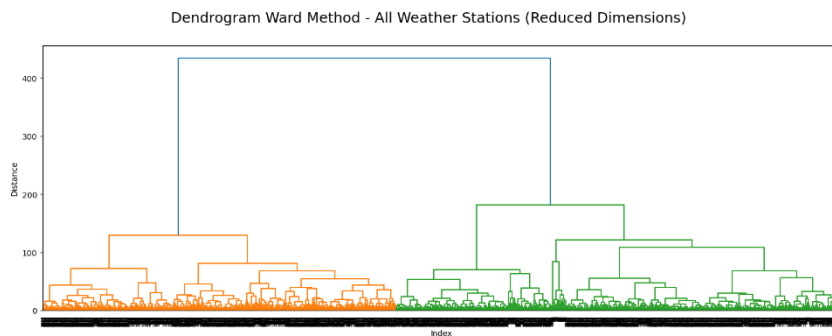
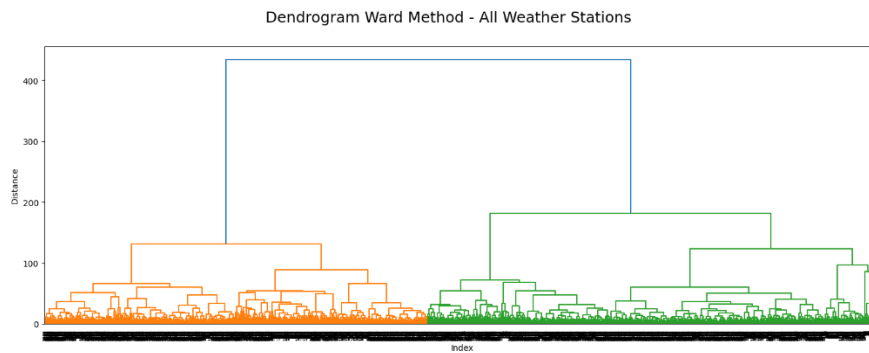


Dendrogram Average Method - All Weather Stations (Reduced to 2 Components)



Ward Method

- The Ward method produces two clusters with similar shape and distribution both before and after applying PCA.
- As with the unscaled data, this method produces two distinct clusters, with one increasing in size while the other decreases as the number of components is reduced.

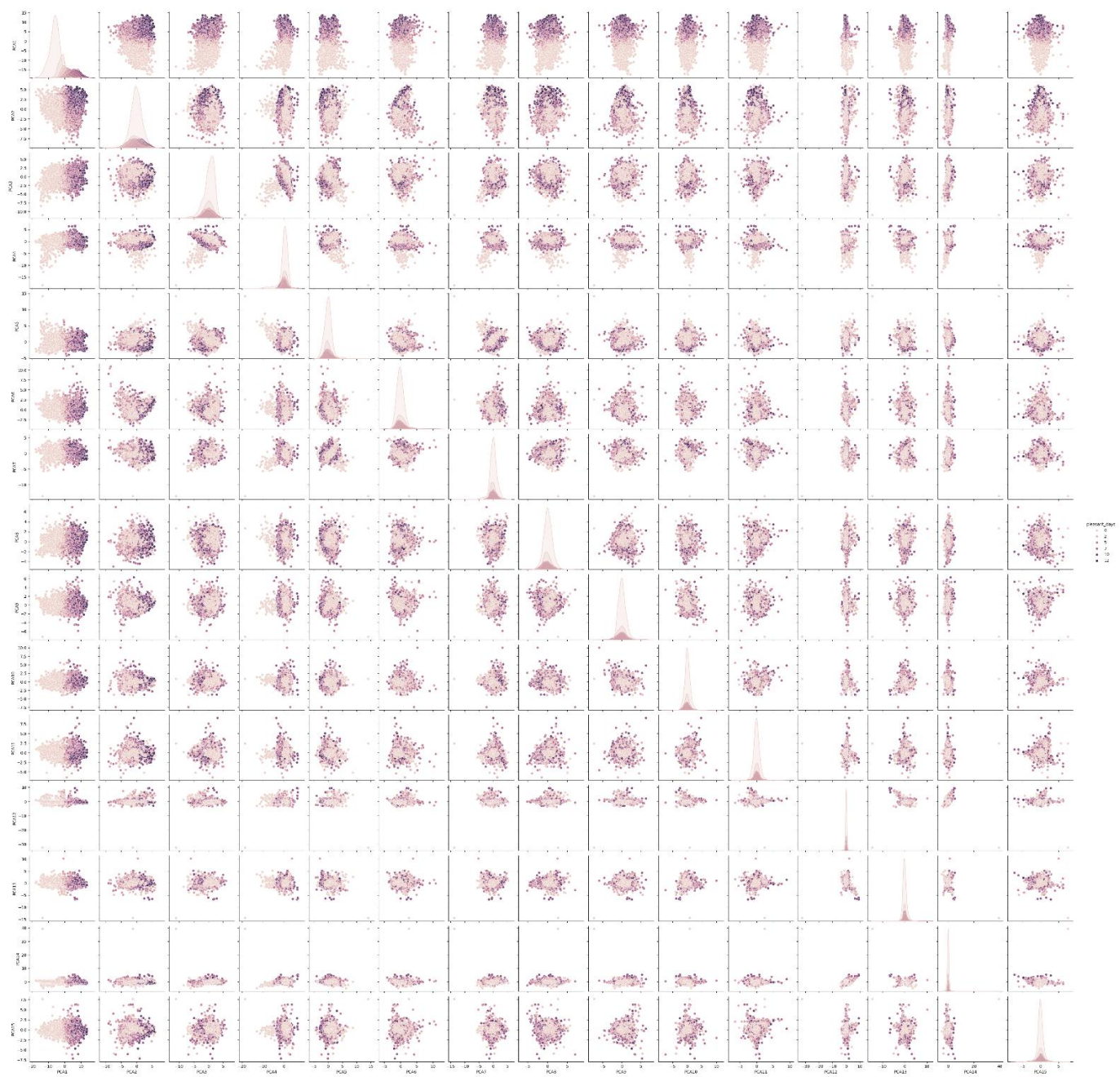


Pair-plot with Pleasant Weather Answers

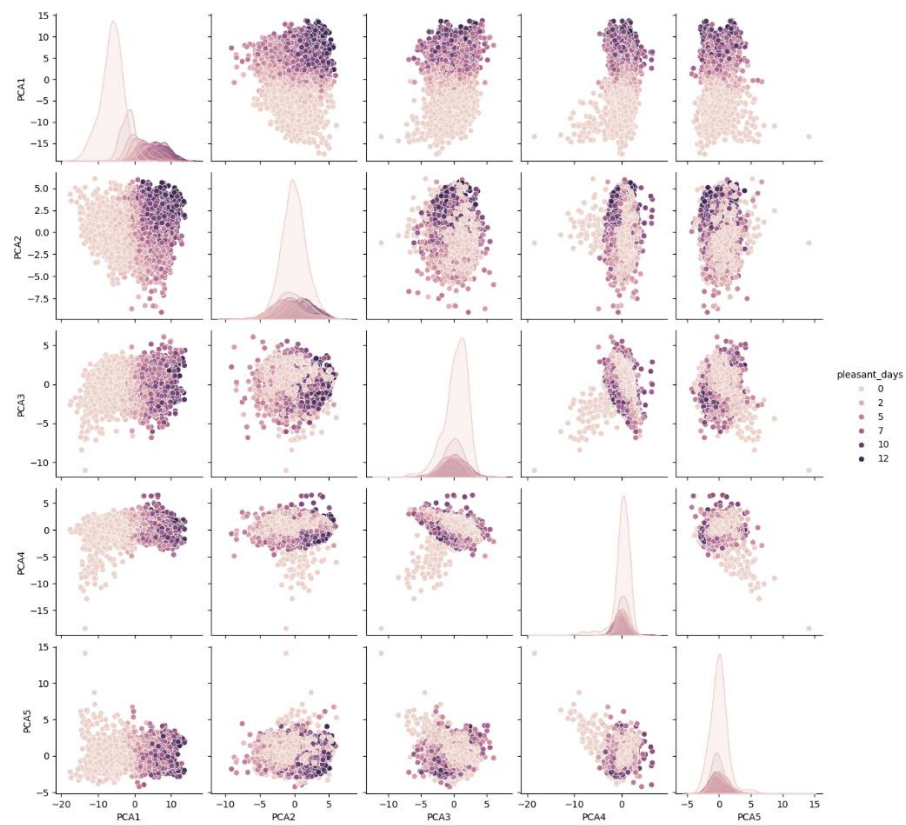
Observations

- The pair-plot with 15 components shows one component distinctly affecting the number of pleasant weather instances, while the rest of the components do not show as distinct of a separation. This indicates that, even after scaling the data, there is at least one component that significantly contributes to the likelihood of a day being labelled pleasant.
- This component remains present throughout the PCA reductions.

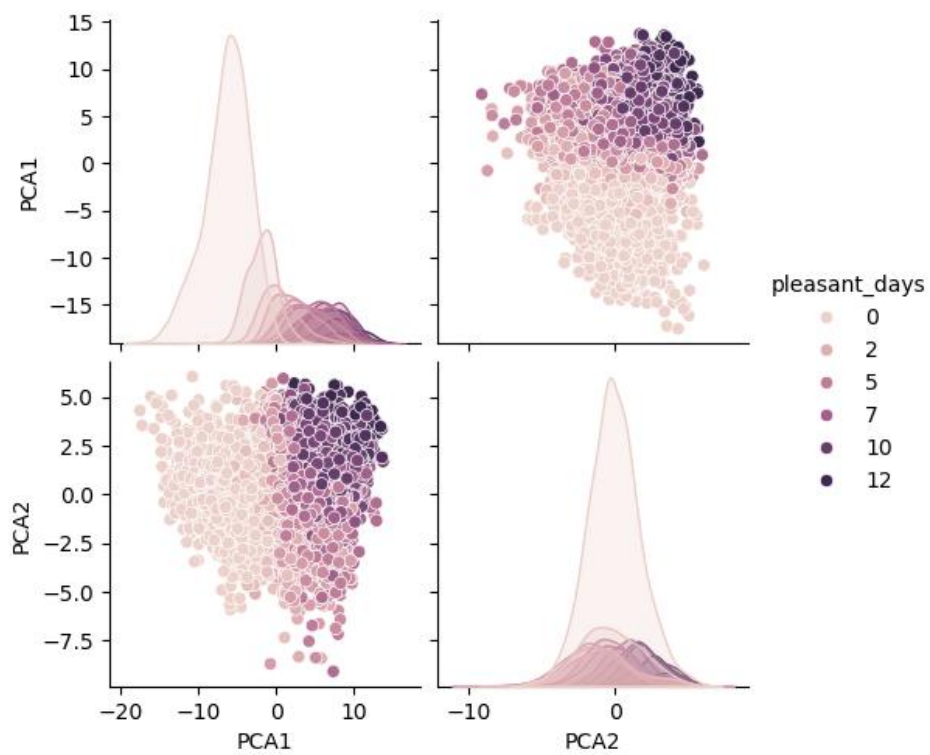
PCA Reduced (15 Components)



PCA Reduced (5 Components)



PCA Reduced (2 Components)



Extracting a single year of data

Data taken from all weather stations over the year 2015.

Weather Data from All Stations, 2015 (Unscaled)

Dendrograms and Pair-plots can be found in the attached Jupyter Notebooks.

Observations

- The Single method produced a single large cluster with multiple colours.
- All other methods produced the same number of clusters with similar shapes and sizes, with those produced by the Ward method appearing flatter than the others.
- In the data reduced to 15 components, PCA1 appears to match most strongly with Pleasant Weather answers in the pair-plot. This component remains even when the data is reduced to 5 and 2 components. At this point, we can assume that this is due to the data being unscaled and this particular component carrying a bigger weight.

Weather Data from All Stations, 2015 (Scaled)

Dendrograms and Pair-plots can be found in the attached Jupyter Notebooks.

Observations

- The Single method produces a similar chain-shaped cluster as it did with other datasets.
- As the number of components is reduced, a second cluster appears to form.
- The Complete method produces 2 large and 2 small clusters with the complete set of data. With 15 components, we can see 3 larger clusters and 1 very small one, while as the number of components decreases, the distribution develops into two even clusters.
- The Average method produces clusters with a similar shape and size distribution to the Single method with the full set of data. As the number of components decreases, the size and shape of the clusters become more uniform.
- As with the other datasets, the Ward method produced two similar-sized and shaped clusters.
- In the pair-plot with 15 components, PCA1 appears to have a stronger influence on pleasant weather compared to the other components. This indicates that, even with scaling, there is at least one component in the weather data that significantly affects the likelihood of a day being labelled pleasant or unpleasant.

Extracting a single weather station

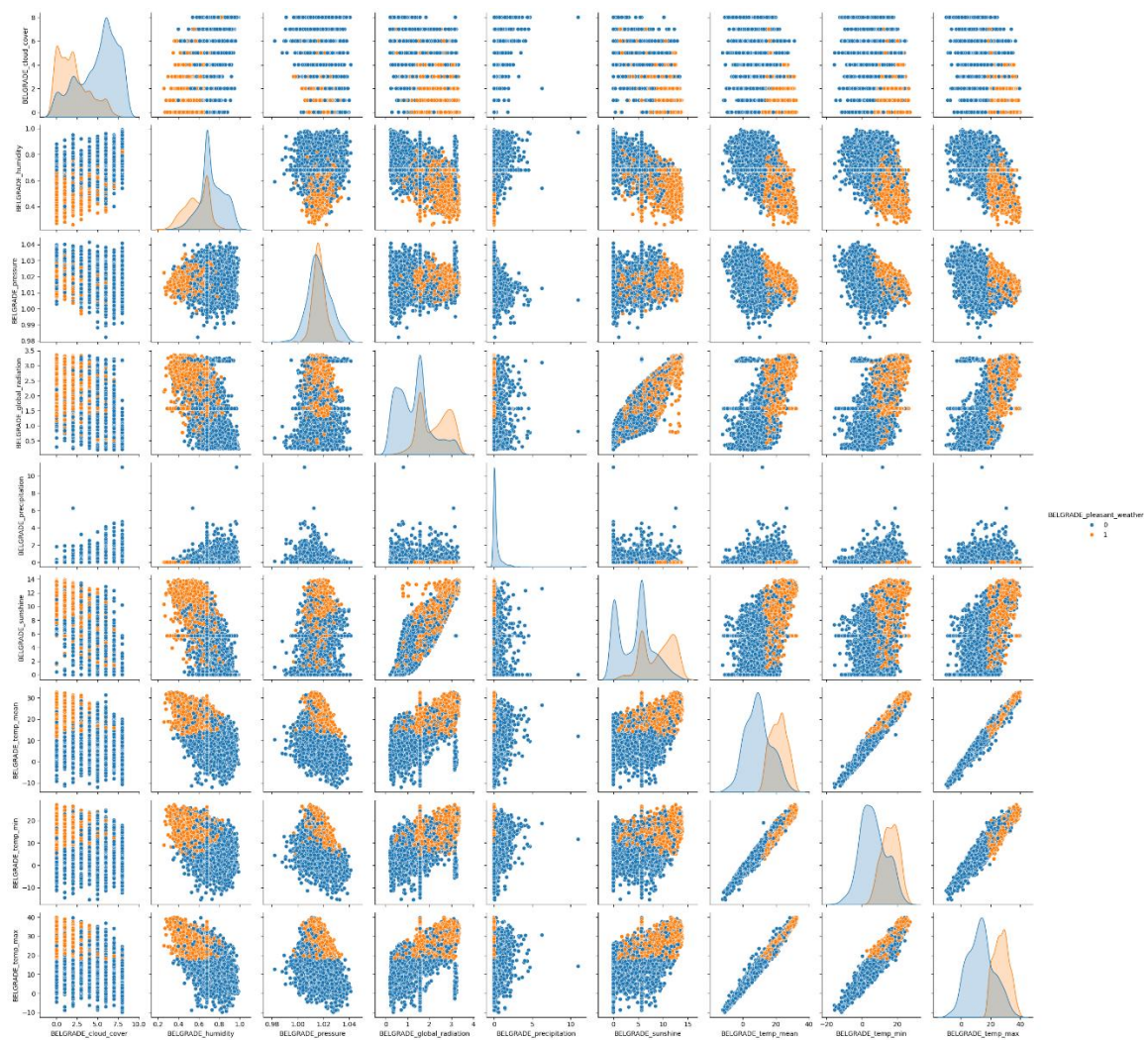
Data taken from Belgrade weather station between 2010 and 2020, using all variables measured.

Weather Data from Belgrade, 2010-2020 (Unscaled)

Dendrograms and Pair-plots can be found in the attached Jupyter Notebooks.

Observations

- The Complete and Average methods produced more clusters as the components were reduced.
- The plot comparing the complete unscaled data with the pleasant weather answers shows a clear overlap between pleasant weather and higher temperatures (minimum, maximum and mean), higher levels of sunshine, as well as lower levels of humidity, precipitation and cloud cover.



Weather Data from Belgrade, 2010-2020 (scaled)

Dendrograms and Pair-plots can be found in the attached Jupyter Notebooks.

Observations

- With the scaled data, the size of the clusters produced by the Ward method increases even as the number of components decreases.
- The Single method appears to produce a small, second cluster when the data is reduced to a single component.
- The complete scaled data produces one very large cluster next to a much smaller one using the Average method. With the data reduced to a single component, the Average method produces two even-sized clusters, with a shape similar to the clusters produced by the Ward method.
- The plots comparing the scaled data with the pleasant weather data show a clearer separation of pleasant and unpleasant weather days, even when the data is reduced to a single component.

Weather Data from Valentia, 2010-2020 (Unscaled)

Dendrograms and Pair-plots can be found in the attached Jupyter Notebooks.

Observations

- Valentia did not have any recorded pleasant weather days between 2010 and 2020, therefore it was not possible to compare the two datasets.
- The Complete and Average methods produced a large, single cluster with several nested layers.
- The Ward method produced one large and one smaller cluster with all components, as well as PCA reductions.

Weather Data from Valentia, 2010-2020 (scaled)

Dendrograms and Pair-plots can be found in the attached Jupyter Notebooks.

- The Complete method produced a single cluster until the data was reduced to 2 components, at which point it produced 3 clusters.
- The Average method produced a long, single cluster (similar to that produced by the Single method), however with only one component, a very small second cluster appears.
- The Ward method produced 2 small clusters and one larger one with the complete scaled data. With each PCA reduction, the Ward method produced one small and one large cluster.

Conclusions

- The single clustering method does not produce meaningful insights with any of the datasets used.
- For data taken from a single location, the complete clustering method produces more detailed insights as more components are removed through PCA.
- With data from all stations, the average clustering method produces more detailed clusters as the number of components in the data is reduced. **This suggests that the average method would work well for identifying broader regional trends.**
- For data taken from multiple locations over a longer period of time, the Ward method provides two distinct clusters with each level of component reduction.
- At least **one component of the weather data strongly corresponds with pleasant weather** across all stations throughout the decade.
- Based on the pair-plot using Belgrade data, we can observe that this component is likely to be **temperature** (mean, maximum or minimum) or **sunshine**.
- Even after scaling, this single component closely matches the distribution of pleasant weather days, both in individual locations and across the region.
- We can therefore assume that the Ward method is best for classifying weather data from a wider region based on the occurrence of this particular component.